Devesh **Walalawalkar**

COMPUTER VISION RESEARCHER & ENGINEER

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Summary

Computer Vision Researcher and Engineer experienced in building and scaling AI based products for a diverse set of applications. Experience working at wide range of companies, from stealth startups to big tech. Has a diverse skill set that ranges from conducting fundamental AI research, including publishing at top AI conferences to deployment of SOTA computer vision tech into real-world products. Strong **undergraduate background in electronics hardware** and **graduate AI research experience** facilitate critical contribution to all areas of a AI based product's research-to-deployment cycle.

Work Experience

Ultron AI
Los Angeles, CA, USA

RESEARCH ENGINEER - COMPUTER VISION — *Oct 2025 - Ongoing*

- Developed a custom object detection architecture with real-time inference constraints for retail AI use-cases, involving automated checkout and out-of-stock detection, which enabled edge AI system deployment
- Deployed custom architecture on edge AI platforms using LiteRT, performed model optimization steps such as pruning and quantization, which enhanced device compatibility and reduced real-time system latency.

Flawless AI
Los Angeles, CA, USA

AI RESEARCH ENGINEER — *Aug. 2022 - Sept. 2025*

- Conducted research and deployment of movie character facial analysis tech as part of the core founding team. This included building scalable R&D frameworks from zero for modules such as Detection, Identification, Motion tracking, Landmarks and Segmentation of character faces.
- Developed a fully automated and scalable labeling pipeline for face parsing, including facial occlusions and beard. This involved the use of foundational models such as SAM2.1 and Radiov3, with a bootstrapping mechanism using initial coarse labels.
- Developed an efficient data sampling algorithm for finetuning a generic Neural Rendering model on character specific videos/shots, reducing the dependency on large amount of finetuning samples.
- Led winning teams at 3 major internal hackathons aimed at prototyping innovative movie AI product tech.
- Mentored interns for various research projects, leading to improvements in character facial analysis performance within core product.

Honeywell
Pittsburgh, PA, USA

ADVANCED COMPUTER VISION RESEARCHER — *Jan. 2020 - July. 2022*

- Developed core AI-based robotics perception technology for warehouse automation systems, including robotic arm depalletization and smart sortation systems, improving warehouse operational efficiency.
- Researched simulation software-based synthetic data capture to enhance an AI-based 2D package segmentation model's accuracy across diverse package designs and placements
- Facilitated seminars and research talks to distribute AI specialty knowledge throughout the organization, enhancing team competencies in AI.
- Deployed large AI perception models on edge AI devices using Nvidia Jetson and Qualcomm platforms, facilitating on-prem deployment.

Biometrics Lab, Carnegie Mellon University
Pittsburgh, PA, USA

COMPUTER VISION RESEARCH LEAD — *June. 2018 - Dec. 2020*

- Led a team of deep learning researchers to develop an iOS application for driver drowsiness detection, enhancing driver safety by enabling driver alertness monitoring through face detection and landmarking, face pose estimation, and eye and mouth closure detection using the iPhone's limited processing capacity.
- Conceptualized and managed the creation of a proprietary dataset having more than 300 subjects (as part of CMU research study) to train Deep Learning models for Human sleepy face detection.
- Led research projects for the US Department of Defense, focusing on object detection model compression and deployment for AV drones, enabling real-time AI based detection capabilities for drones.

Education

Carnegie Mellon University
Pittsburgh, PA, USA

M.S. IN ELECTRICAL AND COMPUTER ENGINEERING — *Jan. 2018 - May. 2019*

- Focus on Machine Learning, Artificial Intelligence, Computer Vision and Robotics department courses

Select Publications

- **First Author** - "Online Ensemble Model Compression using Knowledge Distillation." In ***ECCV 2020*** (Paper link)
- **First Author** - "VideoClusterNet: Self-supervised and Adaptive Face Clustering for Videos." In ***ECCV 2024***. (Paper link)
- **Best Paper Award** - "Medal: Accurate and Robust Deep Active learning for Medical Image Analysis." In ***ICMLA 2018*** (Paper link)
- **Awarded U.S. Patent** - Method for compressing an AI based Object Detection Model for deployment on resource limited devices (link)

Skills

- **Research** for Computer Vision tasks including image analysis and generation, notably Facial Analysis (Detection, Identification, Landmarks), Semantic Segmentation, Generative Video Editing, Active Learning, Medical Image Analysis
- **Efficient Deployment** of Computer Vision tech for diverse domains including Robotics Perception, Movie/TV content, Video Surveillance etc.
- **Software Stack**: Python3, Pytorch, Numpy, Tensorflow, OpenCV, Apple CoreML, Git, HDF5 Database Management